Challenge 3: User Experience for improving values in Large Language Models (LLM) (Team 10)

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Introduction:

Over the past few weeks, our multidisciplinary team, existing of Helena Mossel (Experience Design), Amna Strojil (HCI Design), Rosa van Wershoven (Inclusive Design) and Stefanie Zins (Social Design), has been working on challenge 3, given by Bureau Moeilijke Dingen. Our collective motivation is driven by a shared ambition to shape the future, particularly through enhancing Large Language Models (LLMs) like ChatGPT while highlighting human values in technological developments.

The challenge introduced by Bureau Moeilijke Dingen is to "Design a concept where participants are presented with hot topic items and devise a method that incorporates a democratic process for discussing, critiquing, and considering these topics, with the goal of the output offering insights on how to improve Large Language Models (LLMs)". The ultimate goal is to generate insights that contribute to the refinement of LLMs, ensuring they align more closely with human values. The challenge specified that the implementation must be practical, scalable, and capable of aiding OpenAI in better aligning with its values, while also requiring a clear timeline presentation.

Although the challenge is quite complex and multifaceted, it is important for design students to engage with this topic since it is about enhancing AI to align with human values. Our team has chosen to narrow our focus to the role of LLMs within design processes. This decision is not only influenced by the relevance of this topic but also by our personal interests as design students. Understanding the implications of LLMs in design processes is crucial for several reasons. Firstly, as designers, we are at the forefront of incorporating emerging technologies into our practices. LLMs offer powerful tools that can streamline and add to the creative process. However, the integration raises ethical questions regarding autonomy, bias and human-values in design.

Secondly, exploring LLMs in design processes presents an opportunity to address the broader topic on AI ethics and human-centered design. By researching how these models are used within design workflows, we can contribute to the ongoing discussion on responsible AI development. This is important for shaping the future of AI. As design students, we are not only users but also future creators of AI-driven systems. By understanding the implications of LLMs and their advantages, we can better teach ourselves and future generations of designers to navigate the ethical and other challenges posed by AI.

Our overall goal in this challenge has two sides: to equip Bureau Moeilijke Dingen with a methodology for researching LLM's alignment with human values and to gain valuable insights from our collective experience in applying new knowledge from this course. We have been eager to learn from our peers, understanding and taking a critical look at how they approached similar or different challenges and using their insights to enhance our own skills as UX designers.

UX Challenge:

Approach

Our approach to this challenge was activities-based, focusing on diverging and converging to gather information and narrowing down the data collected. These iterative activities consisted of various tasks done both individually and as group that were aimed to either collect data or narrow down important insights. Before beginning this process, however, we identified a focus for the project, which was the use of LLMs in the design process, leading us to conduct research within the domain of design education. AI usage is a hot topic within design and design education and as design students, we were also able to empathise with this focus area, hence our reasoning for choosing it.

To begin with, we conducted exploratory interviews in which design students were asked 5 questions relating to AI. These questions addressed how they use AI in their design processes, risks/benefits to AI, and how they believe AI will play a role in design in the future. For example, some of the questions asked were 'how do you feel when someone uses AI in design?' and 'How do you think AI will play a role in design in the future?' The majority of the responses we received in these interviews indicated that design students saw AI as a tool that could be beneficial in the design process, yet faced many risks, such as losing the ability to create ideas without assistance due to relying on it too heavily. We also asked ChatGPT, likely the most well-known LLM, the same questions. ChatGPT also provided answers indicating both high risk and reward from using AI in design work, yet it was done in a much more objective and impersonal manner. Receiving both types of responses allowed us to gain a broad 2nd perspective understanding of the design students' cumulative user experiences with AI from their perspectives, while comparing them to the generated answers[1].

This activity also led us to using a value-sensitive approach as a next step in the design process to frame which values were at the forefront for design students [2]. Based on the responses received, we identified that their top values are inspiration (using AI as a tool when they feel creatively blocked), (degree of) ownership (wanting to oversee their ideas and processes), support (using AI to enhance their processes), and efficiency (using AI for difficult or tedious tasks). Identifying these values was an important part of the process, as it allowed us to empathize with design students and understand their perspectives, leading to being one step closer to identifying how to better align LLMs to real human values.

The final step in our approach before converging again to propose a solution involved background research on 4 relevant topics: Large Language Models (LLMs), ethics of AI, data collection with AI, and the role of AI in design processes. Firstly, gaining understanding on LLMs was necessary in being able to identify a clearly defined solution. LLMs are AI models trained on a large data set that can generate human-like text, yet they still possess many limitations and often have ethical issues [3]. There are currently many guidelines that have been developed to ethically mediate the relationship between AI models and human usage of them, often highlighting the importance of transparency, justice, responsibility, privacy, and autonomy [4]. Ethics and morality remain an important theme in research and development of AI models.

As LLMs are trained on large data sets, they continuously interact with large quantities of data. Data collection is necessary for machine learning and has a direct impact on the performance of the LLM, but comes with its own set of ethical problems, particularly regarding privacy and the storage of personal data. LLMs can also, however, be a useful tool in the processing and analyzing of data, leading to quickly uncovering insights and collecting information that can be used in design processes [5]. AI can be used to aid designers in various steps of the design process, such as generating prototypes, analysing user interactions, checking designs for accessibility, and many other tasks [6]. This shows an opportunity to elevate the design process, enhancing creativity, efficiency, and idea generation. Seeing the potential of LLM usage making a positive impact in the design process, while also acknowledging that there is a fine line between appropriate and inappropriate use of it led us to a design solution that aims to align the two.

Solution

Studying values serves as a pathway to understanding needs and behavior, as it allows insight into the factors that influence behavior and are influenced by needs [7]. Therefore, we focused on observing practices and retrieving in-depth insights and values for utilizing OpenAI in designers' workflow. The method (Figure 1) is a one-day workshop with industrial design students. This target group has been chosen to perform the workshop with, because this is the generation that is currently exposed a lot to the use of AI as tools in education [8]. They will be given a design brief for which they have to complete all steps of the design thinking process [9]. However, the participants will be divided into three groups (Figure 2): group A will be restricted to not use any LLM at all. Group B is free to choose when they want to use an LLM, and group C is required to use an LLM at least once in every stage of the design thinking process. All groups will be given a camera and process book to document their process and a blank poster on which they can eventually present their results. The method will be a single-blind study, so none of the groups will know what category the other groups fall into. In turn, at the end, the participants can classify the posters into the extent they think AI has been used to come up with the solution. After the actual results have been revealed, a focus group will be held with participants from various groups sitting together. This is inspired by the Empathic Handover method by Smeenk [10], as this theory describes how empathy can be transferred to team members that did not meet the user in person. In a similar way, participants from the various groups can share their experiences with each other, giving the others insights on their process, without them actually having been there. These discussions will be led by the researchers to assure that other's insights could be related to another participant's own anecdotes and experiences. The focus of the discussions will be on investigating how LLMs influence the values that had previously been investigated by us from the interviews. The participants eventually propose a design process where AI is implemented, utilizing its qualities efficiently at the right design phases.

Discussion

Various points of discussion arose from this solution. This approach is based on qualitative data to gain insights on improving the UX of OpenAI for designers. However, OpenAI focuses a lot on quantitative data, because whereas you can get most in-depth insights from qualitative data, quantitative data is easier to implement on a bigger scale. On top of that, LLMs work on large quantitative data sets, as that is what is used to train them and therefore, qualitative data is not enough. Our current solution was aimed to be implemented with 27 participants, which is a small sample size for such a big organization. Therefore, it needed to be considered how this could be quantified and upscaled. We propose that initial insights and values can be gathered from a one-day workshop and in turn validate and generalize these insights through a more quantitative approach (e.g., a questionnaire).

Another potential solution involves transitioning the workshop to an online platform (Figure 3), offering the ability to accommodate a larger participant sample. This new approach could keep the random allocation to groups A, B, or C, but adapt the activities into tasks completed with or without the assistance of AI, depending on the assigned group. Subsequently, the results of these tasks could be classified as “Done with AI” or “Not Done with AI,” integrated into a gamified format to collect participant opinions.

Following this, multiple chat rooms would be established to facilitate discussions, each randomly assigned with participants from the three different groups. In these chats, an LLM would serve as a moderator, generating conversation prompts and summarizing participant inputs. Participants would have the opportunity to endorse or change the summaries, fostering collaborative reflection and brainstorming between AI and humans.

Through this method, we aim to clarify the role of AI in inspiring and supporting users, leveraging the collective insights of participants to generate valuable outcomes. This approach not only harnesses the accessibility and scalability of online platforms but also fosters inclusive and interactive discussions, enriching the workshop experience.

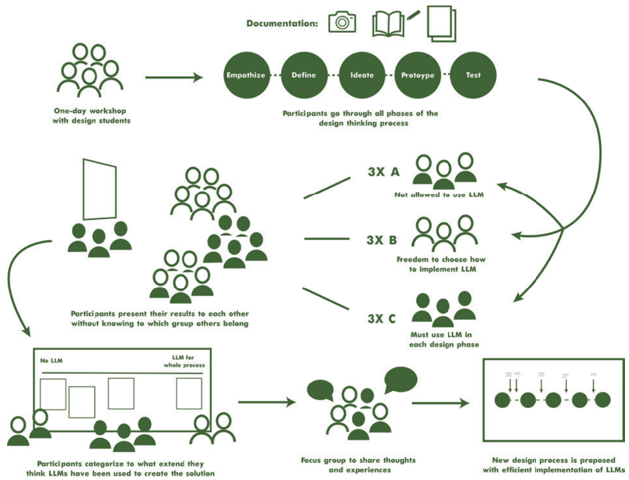

Figure 1: Workshop set-up

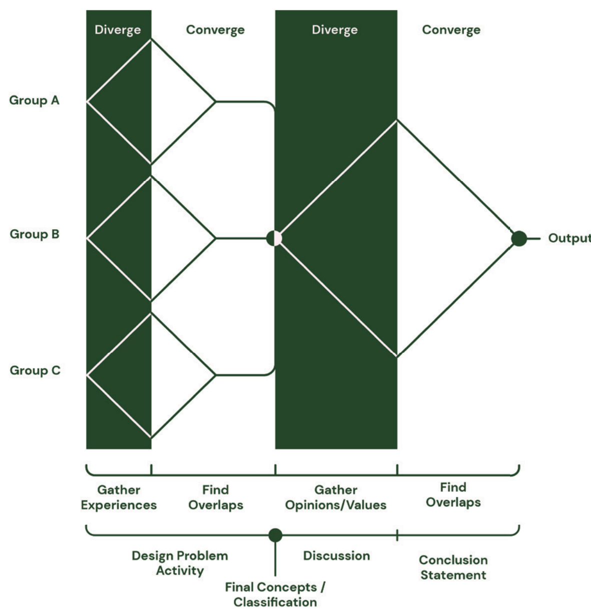

Figure 2: Activity Flow

Furthermore, as of now, this approach has been focused on improving OpenAI for designers, but we have not considered generalizing it for other fields. However, we argue that a similar approach can be applied to other fields as well. Generalizing this method, people can be asked to perform a certain assignment from their field of expertise. They can then be divided into three groups as mentioned in the original approach. On top of that, a focus group with participants from groups A, B and C can always be executed to compare experiences and results.

Finally, we acknowledged the limitation of various degrees of experience with LLMs, but we did not integrate a solution for it. However, we do realize that it is a very interesting approach to investigate how even people who are inexperienced with LLMs can implement it as a tool to benefit their workflow. For this, we do not think that much on the approach itself has to change, but in the consideration of how to divide participants into the different groups. We think that if people who are inexperienced with LLMs are put in a group where they are required to use LLMs in every phase of the process, they will learn about the possibilities there are with it and based on that form their opinion.

Limitations

As with any design solution, we are faced with several limitations relating to both the system as a whole, as well as the individual components of the solution. As previously mentioned, various levels of LLM knowledge may affect the flow of the workshop. Since the workshop is a one day event, mediation of this may be difficult due to limited time, likely interfering with the participants’ ability to share and apply knowledge of LLMs. It is not fair to assume that all participants will have experience working with LLMs, therefore, it should be noted that these different levels of knowledge could interrupt the workshop’s structure The depth of both knowledge sharing and data gathering have the risk of remaining relatively superficial, while also raising concerns regarding transparency. Participants may be concerned about the transparency of the activity itself, as it is done in a single-blind approach, and how the data they provide is stored and processed. Ethical gathering, storing, and

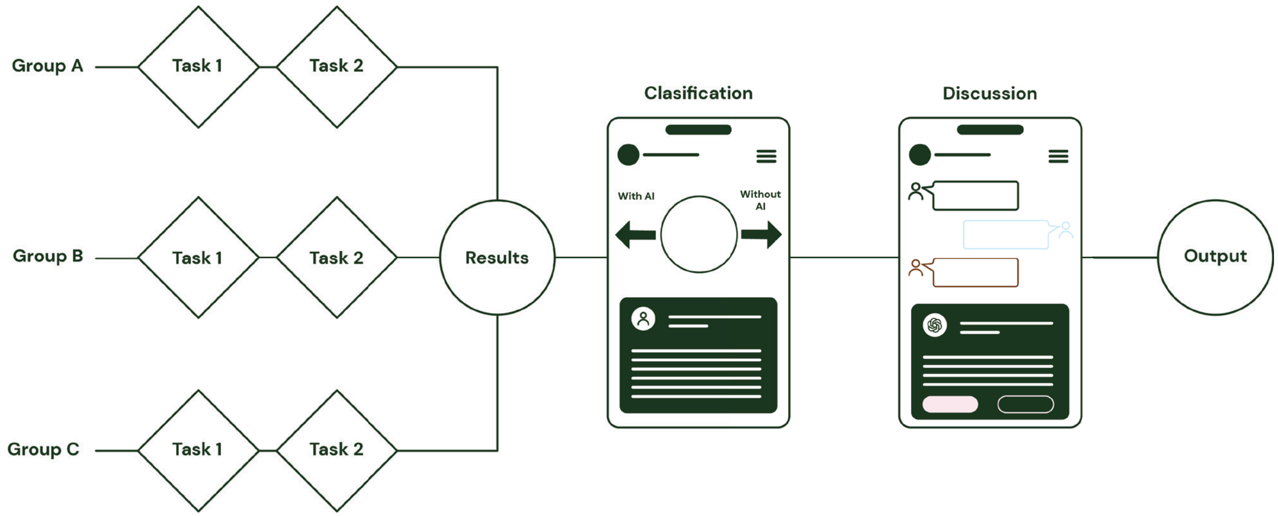

Figure 3: Possible improvement of Solution 1

processing of data will be required and participants must be clearly informed about this, due to the possible risks involved. The limited time may also play a role in the motivation of the participants, possibly affecting their willingness to learn and engage in the design process. The length of the workshop may incentivise more participants to join, as it only requires committing to a single day, yet this may limit the amount and depth of data and insights that can be gathered.

In relation the overall impact of the activity, we are also limited by the technological capabilities of LLMs, as well as the impact that they have on the environment. Pairing the limitations of the LLMS themselves with the participants' possible limited knowledge can impact the ability of the design process to unfold. The LLMs are also trained on very large data sets and require significant amounts of energy to both be trained and run, posing an ethical issue on the environmental impact that such a workshop may have [11]. Overall, while our design solution has the potential to gather useful data in the use of LLMs in the design process, it still faces limitations of various scales.

Next Steps

For our next steps we will be iterating upon our solution based on the feedback we received. In turn, we will start looking into validating our improved solution by first recruiting the 27 participants and organizing a pilot test. The insights of this pilot test will be useful to iterate again upon the solution. More specifically, we want to investigate how the results from this test could be generalized and applied on a bigger scale. From there we will execute the actual test by inviting professionals in the field of design to also join the workshop. This output will also be used in a more quantitative approach to validate the results on a bigger scale. Once this approach has been deemed to be successful for generating insights for the development of LLMs in the field of design, the approach will be applied in other fields of expertise. In the end, these insights will then be used for the development of LLMs to improve its UX in the work field.

Review of Challenge 1 and Challenge 2:

Challenge 1 (Dopper):

Team 1, Team 2, Team 3, and Team 4, all presented their own approach addressing Challenge 1 problem for the client, Dopper. The Challenge was to create a distinctive user experience with one of the user stories to get a Dopper as a starting point, that results in Dopper meeting their goal to build long-term consumer relationships and increase revenue.

Team 1 had a structured analysis, employing a customer journey to dissect the problem and propose a scalable solution in line with Dopper's core values of sustainability. Their approach lacked a usage scenario and focused solely on new customers, leaving room for curiosity about engaging existing customers or customers that had received a Dopper has a gift. On the other hand, Team 2 implemented a scenario that made their solution easier to understand and provided a more detailed view of the experience. They had thought of multiple possibilities in which the user would find himself and how they could interact with the device. Yet their proposal of a soda dispenser seemed out of sync with Dopper's mission of promoting water consumption, not relevant to the needs of the main users.

Team 3 took a user-centric approach, conducting interviews to understand the product's usage context better, leading to a tangible solution aligned with Dopper's existing infrastructure. However, their oversight in not depicting a daily life scenario left a gap in fully realizing the potential of their solution beyond seasonal use during festivals and other events. Meanwhile, Team 4 adopted a systemic perspective, delving into the customer journey and return policy to suggest a mechanism for reconnection through the recycling of damaged pieces. Though promising, the extended timeframe for results made it less practical for Dopper's immediate needs, given the durability of their products.

Despite their differences, each team's presentation contributed interesting insights, highlighting the complexity of reconnecting, or keeping the connection with customers. However, this challenge serves as an example to show the importance of UX in creating and keeping those connections.

All in all, the groups provided solutions that used a Dopper as an instigator or facilitator of an experience. The solutions were different, but all circled around the Dopper bottle. The rest of the teams looked at the experience during the use of the bottle. Contrarily, team 4 saw a possibility for a systemic touchpoint at the end of the cycle of life of the bottle.

Looking at the paper from Brand, R. and Rocchi, S [12] has aspects from the two first economical paradigms. For instance, the fact that the experience is connected to the purchasing of a product relates back to the Industrial Paradigm. In addition, the branding of the company takes a closer look at the customer and focuses on the pre purchase, the purchase and the post-purchase experience which has a relation with to the Experience Paradigm. However, this challenge had a potential to be looked at from the transformation economy paradigm perspective. It might be interesting to see how the more the human perspective could be considered when designing for this challenge as sustainability is a key factor of this challenge.

Challenge 2 (CM.com)

For Challenge 2, each team offered unique insights into how companies should create Conversational Interactions with their end-users; that these users love to receive (and to respond to) to build an enduring relationship. They all had varying approaches and methodologies to tackle this problem. Team 5's presentation lacked a clear scenario illustrating user-system interaction and the customer journey, not making clear the significance of the UX for both the company and the user. Nonetheless, their idea of a familiar tone in the experience was an interesting strategy for growing lasting user relationships through the simulation of talking to a friend or someone close. Additionally, implementing transparency regarding clothing recycling emerged as a trust-building mechanism between users and the company.

Team 6 had a similar idea to Team 5's focus on utilizing tone to shape positive experiences but introduced the concept of adaptability, personalizing message tones based on subject severity. Meanwhile, Team 7 delved into user interviews to drive their process, leveraging insights to craft personas, identify needs, and map out touchpoints through a detailed customer journey. Their method, shifting from second-person to third-person, and to first-person perspectives [13], showcased a holistic approach to user empathy, enriching their solution design helping them to stay close to the user and satisfy their needs.

Moreover, Team 8 introduced the Wizard of Oz method, a compelling approach for user testing in user experiences. Their demonstration highlighted an innovative possibility for implementation of an experience for new customers of the NS to help them manage their travels. Collectively, these presentations show diversity of user experience solutions, emphasizing the importance of empathy, transparency, and adaptability in creating meaningful experiences that resonate with users on various levels.

In other words, Challenge 2 was about creating experiences that would foster a lasting connection between a user and a company. The groups ended up with different solutions depending on the company they had chosen. However, they all had an experience based on the users need for support and information. Contrarily to Challenge 1 this Challenge did not have a specific object to facilitate the experience, it relied on digital technologies and communication. The digital environment is different than the material environment, this means that to design a user experience in this environment, it is necessary to understand how users interact with the digital platform, what are their expectations and needs. The teams assigned to this Challenge understood well the interaction of the WhatsApp platform through their research, as it was important to understand the features available to them to provide an experience through this platform. On the other hand, they understood the expectations and needs of the users to have quick and easy access to information which had a good connection to the features of WhatsApp. In other words, to create an experience when designing in a digital space it is important to understand the environment and how the user will interact with it.

The groups' solutions provided examples of approaches to user experiences such as the change of tone depending on the seriousness of the subject, or the familiarity of WhatsApp to communicate. Through these experiences, the user can form a connection with the company and trust, which on a larger scale provides longer and stronger bonds between customer and company.

This challenge also shows aspects of multiple paradigms, but most specifically the knowledge paradigm [12]. It is visible by the possibility to share and communicate personalized data and gather it to improve the system. However, it can be argued that building personal connections with the user is also an aspect of the transformation economy.

General reflection:

In comparing the three challenges of this course, it is clear that each one of them offers a unique set of goals.

Challenge 1 is about Dopper, a company with a clear sustainability plan but limited post-sale customer engagement. The task is to devise innovative user experiences that foster long-term customer relationships and boost revenue. This challenge is quite business-focused, emphasizing the need to align user centered design strategies with tangible goals. Despite its commercial orientation, it offers a case for exploring user journeys and generating creative solutions. The existing user stories provide a good foundation for brainstorming and implementation. Additionally, the visual nature of the Essense company adds an aesthetic dimension to the design process, potentially inspiring visually engaging solutions.

In contrast, challenge 2 shifts its focus to the company CM.com and the goal of business messaging. The challenge is framed within the context of a paradigm shift towards two-way communication, driven by the increased customer expectations. Unlike Essense, CM.com is characterized by a less visually oriented approach, with strong emphasis on business and management perspectives. The task is to create conversational interactions that match with end-users, with the overall goal of sustaining enduring relationships. This challenge needs a good understanding of marketing dynamics and user preferences, which highlights the importance of user centered design strategies.

Challenge 3 represents a less business-oriented focus than previous challenges, it is about delving into design and technology with a research-focused on systems and data-collection with both a less tangible and solution oriented approach than the other challenges.

The challenge is described by Bureau Moeilijke Dingen is described as complex, emphasizing the need for practical problem-solving skills. Unlike the other challenges, which primarily focus on business goals, challenge 3 places a bigger emphasis on the intersection of design, technology and human-centered solutions. It challenges designers to create tangible outcomes in a dynamic and fast-paced setting. Challenge 3 has a relation with the Knowledge economy paradigm [12]. This challenge shows how knowledge is not longer static but is socially constructed, discussed and shared among individuals, leading to a collective understanding which in this case would be a set of values in LLM's. On the other hand, this digital focus, facilitates real-time social dialogue and exchange across geographical boundaries, connecting people globally and enabling collaboration and knowledge sharing on a large scale.

In summary, while each challenge has its own set of complexities, they all highlight the multifaceted nature of design and the diverse skills required to address these design challenges effectively. From business-driven goals to the exploration of conversational interactions and the hands-on application of design principles, these challenges offer valuable opportunities for learning, innovation, and professional growth as a UX designer.

Reflection of literature in industry

An interesting point of reflection and observation is how in Challenge 1, the implementation of the course literature was quite clear, with tools helping the designers throughout the process. However, in the second challenge, the presence of the literature became quite less. By the time we reached the third challenge, both ourselves and others had difficulties integrating a wide variety of course literature into the casus. While some teams attempted to do so, it was noticeably less apparent compared to the other challenges. The implementation of the literature was more visible in the approaches of the teams that took on challenge 3 than in the final results presented.

However, when listening to the stories and feedback from the various companies, it becomes clear that the literature is implemented more than expected. Many of them use tools and methodologies directly connected to the course, which is interesting to see. While it may not always align perfectly with what we've learned, the influence is still noticeable in the approaches and work methods.

Insights on UX

Through our team's engagement with various challenges, we gained valuable insights into the practical application of UX theory in real-world scenarios. Each challenge provided us with a unique perspective on how UX principles can be used to address complex challenges. These experiences showed that the literature we have studied serves as a toolkit for UX designers, enabling us to develop our own methodologies in specific design processes.

One key takeaway was the recognition of the diverse approaches to UX taken by different companies. We observed how each company has its own interpretation of user involvement and uses various methods and theories to simplify challenges. Moreover, we have learned that UX exists of multiple layers, not only involving understanding user needs and experiences but also carefully considering how to gather and use this information effectively.

Furthermore, the challenges highlighted the adaptability of UX frameworks and theories. While frameworks may appear rigid initially, we discovered their flexibility in helping different processes and problems. This adaptability allows UX designers to navigate both structured and exploratory approaches to problem-solving.

Importantly, the challenges provided us with a first experience to UX design in the commercial world. We have seen diverse perspectives and understandings of user experience among different groups, each applying theory in unique ways to address specific problems. Our own challenge, which focused on conceptual and systemic issues rather than traditional customer-centered concerns, offered an interesting opportunity for research-based exploration and improvement of new technologies.

In conclusion, engagement with these challenges expanded our understanding of UX design beyond theoretical frameworks, providing us with practical insights into this application across various industries and contexts.

Weekly logbook:

When Challenge 3 was presented, we immediately sat down together and discussed the topic. This discussion had an explorative nature, we wanted to know a bit about each other, e.g., where our interests lay, what our opinions were on the use of AI. We ended the discussion by focusing on AI in the design process. However, we wanted to have more opinions on the use of AI in design and the values it should have. Therefore, we opted for interviewing our classmates on this topic. We wrote five simple questions that could be asked to multiple students. We also asked questions to ChatGPT to read the Large Language Model's answers to our enquiries. The result of this activity was 11 explorative interviews excluding ChatGPT's interview. This activity was interesting for us, as it helped us have a better understanding of the users and empathize with the values they hold corresponding AI use in creative processes. After, we met to analyze the interviews. The outcome was a list of four values that our participants found important to have when designing. This part of the activities was pf much use as it steered our discussion and therefore the background research. We also focused better on making an activity tailored to those specific values. We then decided to do background research to gather more in-depth information about the subject. During the next session we discussed the outcome of our research and started brainstorming a concept. Finally, we finalized the concept and made it more concrete before presenting it to the client and the rest of the class. This process, based on activities, was an interesting approach to creating a solution. We felt it made us go faster and take quicker decisions based on the activity's result.

On the side, we also saw the presentation of the other groups to get inspired. Even though the challenges were different we found it interesting to observe our classmates' approaches and learn from their mistakes.

Finally, reflecting on the team dynamics, we were lucky to have a team that was able to communicate well throughout the entire process. We were open from the beginning about our learning goals and helped each other when needed. In addition, we learned much from one another, about how we understood UX and how we saw it evolving. The discussions were always intriguing, and we tried to give each other constructive feedback when needed. Overall, it was a successful teamwork, where everyone did their part and respected each other.

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